

About Me

- UCLA BA 1999, back since 2005
- teach soc 173 and m176
 - those classes are theory oriented, this one is methods
- use networks
 - Climbing the Charts book 2012
 - Rossman, Esparza, and Bonacich ASR 2010
 - Rossman and Fisher PNAS 2021
 - Contexts paper
 - working on a methods piece w Stoddard and another simulation

Why networks?

Fundamental rejection of mass society / perfect competition assumptions

Answer many research questions:

- How do ideas flow?
 - homophily vs influence
 - opinion leadership and optimal seeding
- How do resources flow?
- How do diseases flow?
- Broader sociological questions
 - chain migration
- Identity from network
 - community detection
 - status

Data

- Egocentric
 - “network module” of a regular random sample survey
- Whole network
 - sociometric surveys
 - affiliation networks
 - big data

Tools

- Metrics
 - whole network like clustering coefficient
 - node level like centrality
- Graphs
- Simulation
- ergm
- Load R/ RStudio / igraph

Data Formats

- Edge Lists
- Matrices